Colour-blind Series: Deuteranopia

In short

Deuteranopia makes green hard to see, so red and green can look almost the same. Charts and messages that use only red and green won't work for people with deuteranopia. Always add icons, labels, patterns, or words, so colour isn't the only way to show meaning.

What is it for?

Deuteranopia is a form of colour blindness that affects the ability to see green. Like protanopia, it is a type of "red-green colour blindness". Protanopia affects the cells that sense red. Deuteranopia affects the cells that sense green. It is actually the more common of the two, affecting about 1% of men. It's very rare in women.

In practice, the experience is similar. Red and green become very difficult to tell apart, making colour-coded information unreliable. A traffic light style status indicator, a heatmap, or a form validation message relying on red and green will all present the same challenge.

Deuteranopia is part of a broader family of colour vision deficiencies, and this Colour-blind Series looks at each of the main types. Each type affects a different part of the colour range, but they all teach the same lesson: never rely on colour alone to share something important.

How to support it

The key principle is that colour should always be paired with another visual cue.

- Designers can add icons, labels, or patterns next to colour to show meaning. They can also use a colour blind simulation tool, such as Web Disability Simulator for Chrome, to check designs before they are built.

- Developers can ensure interactive states like errors and warnings are distinguishable by more than just colour (for example, by using icons or supporting text).
- Content writers can make sure any meaning shown through colour is also stated in words.

Useful links

- Web Content Accessibility Guidelines (WCAG) 1.4.1: Use of Colour
- Colour Blindness Simulator (Coblis)
- Colour and Accessibility (WebAIM)

Abbreviations

The full form of each abbreviation used in this document.

WCAG: Web Content Accessibility Guidelines